

Assignment-2: Identify a real-world application for both parallel computing and networked systems. Explain How these technologies are used and why they are important in that context.

⇒ Parallel Computing Example:

Weather Forecasting

when meteorologists predict the weather, they use complex models that simulate atmospheric conditions.

These simulations require enormous computational power. To speed up these calculations, supercomputers like those used in weather forecasting employ parallel computing. In this system, many processors work simultaneously, each handling a portion of the computation. For example, one processor might calculate temperature change in one region, while another handles wind speed in another region. By splitting the workload, the system can deliver weather forecasts in a fraction of the time it would take a single processor.

How it works:

Data collection: A large amount of data from various sources (satellite images, weather stations, sensors) is gathered to represent the atmospheric conditions.

Task Division: The weather prediction model is divided into smaller tasks (e.g., temperature calculation, wind patterns, cloud formations). Each task represents a specific region or aspect of the simulation.

Parallel Processing: Multiple processors (or cores) process these tasks simultaneously. Each processor works independently on its assigned task but in parallel with others.

Data Synchronization: The results from different processors are continuously synchronized to ensure that the simulation work together cohesively. For example, if one processor updates the wind speed, others will adjust the temperature and pressure accordingly.

Aggregation: After the simulation is complete, the data from all processors is aggregated to produce a final weather forecast.

Output Generation: The final results, which include predictions for temperature, precipitation, and other weather phenomena, are provided to meteorologists, who can then issue weather reports or warnings.

Why it is Important:

Speed: Reduce the time needed to process large datasets, allowing for faster forecast.

Accuracy: Enables the use of more detailed simulations, leading to more accurate predictions.

Scalability: Allows for the handling of more complex models and larger datasets as computational power grows.

Real-Time Predictions: Supports the delivery of up-to-date forecasts, which is vital for timely weather alerts and safety.

Networked Systems Example:

Online Multiplayer Video Games

In Multiplayer online games like Fortnite, players connect to servers through the internet to interact with each other in real-time. These games are an example of

networked systems because they involve many computers (Player's systems) communicating with central servers over a network. The servers coordinate the game state, ensuring all players see the same world and can interact with each other. Networked systems allow players to participate in the game state, ensuring all players see the same world and can interact with each other. Networked systems allow players to participate in the game from different locations, but the system must handle latency, data synchronization, and traffic to ensure a smooth gaming experience.

How it works:

Client-Server Model: Player's devices (clients) connect to a central server (or a distributed set of servers) via the internet. The server manages game world and ensures that all players actions are synchronized.

Data Transmission: When a player performs an action (e.g., moves, shoots), their device sends this information to the server over the internet. The server processes the action and updates the game state.

Game State Updates: The server sends the updated game state (e.g., position of all players, changes in the environment) back to all connected clients. This ensures that everyone sees the game world in real-time.

Latency Management: Networked systems must manage latency (the delay between sending and receiving data) by using techniques like prediction and interpolation.

to ensure smooth gameplay, even with delays in data transmission.

Consistency: The system ensures that all players experience the same game world. The server resolves any conflicts, such as when two players try to occupy the same space at the same time.

Persistence: Many games maintain a persistent world where actions taken by players (like leveling up or completing quests) are saved and reflected across sessions. The server manages this persistent data storage.

Content Delivery: Networked systems can also be used to deliver updates, patches, or new contents to players in real time.

Why it is Important:

Global Connectivity: Enables players from around the world to interact in the same game, creating a global community.

Real-Time Interaction: Ensures players actions and changes in the game world are reflected instantly, providing a seamless experience.

Scalability: Allows games to support a large number of players simultaneously without compromising performance.

Persistence: Ensures continuity in the game world, where progress is saved across different sessions players experience consistent.

Dynamic Content: Facilitates the delivery of fresh content (like updates or new features) to players in real-time, keeping the game experience engaging.

Both parallel computing and networked systems are foundational to modern technology and help power some of the most complex and widely used applications in fields ranging from scientific research to entertainment.